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CS929 coursework2

I am typing this paragraph because my assignment was originally written by hand, and I need to provide a digital version for submission.

Question 1

I \ II	A	B
1	1, 1, 1	3, 2, 2
2	2, 0, 1	0, 1, 1

Player III chooses strategy 1

I \ II	A	B
1	0, 1, 2	2, 0, 1
2	4, 4, 4	0, 1, 0

Player III chooses strategy 2

(a) Dominated Strategy: A strategy is dominated if there exists another strategy that always gives a player a high payoff, regardless of what the other players do.

For Player I, Player II choses
- A ⇒ Player III choses 1 ⇒ Payoff of Player I choses { 1 is 3 (win!!), 2 is 0 }
- B ⇒ Player III choses 1 ⇒ Payoff of Player I chooses { 1 is 1, 2 is 2 (win!!) }

For Player II, Player I choses
- 1 ⇒ Player III chooses 1 ⇒ Payoff of Player II choses { A is 1, B is 2 (win!!) }
- 2 ⇒ Player III choose 2 ⇒ Payoff of Player II chooses { A is 4 (win!!), B is 0 }

For Player III, Player I chooses
- 1 ⇒ Player II chose B ⇒ Payoff of Player III choses { 1 is 2 (win!), 2 is 0 }
- 2 ⇒ Player II chose A ⇒ Payoff of Player III choses { 1 is 0, 2 is 4 (win!!) }

Answer: No Player has any dominated strategy

Question 1.

I \ II	A	B
1	1, 1, 1	3, 2, 2
2	2, 0, 1	0, 1, 1

Player III choose strategy 1

I \ II	A	B
1	0, 1, 2	2, 0, 1
2	4, 4, 4	0, 1, 0

Player III choose strategy 2.

(b) Pure Nash Equilibrium = A strategy profile where no player can increase their payoff by changing their strategy.

Case 1 : I = 1 II = B III = 1 ⇒ Payoff = (3, 2, 2)

If
- Player 1 change to 2 ⇒ Payoff = (0, 1, 1) No!! 3>0
- Player 2 change to A ⇒ Payoff = (1, 1, 1) No!! 2>1
- Player 3 change to 2 ⇒ Payoff = (2, 0, 1) No!! 2>1

So (3, 2, 2) is pure NE!!

Case 2 : I = 2, II = A, III = 2 ⇒ Payoff = (4, 4, 4)

If
- Player 1 change to 1 ⇒ Payoff = (0, 1, 2) No!! 4>0
- Player 2 change to B ⇒ Payoff = (0, 1, 1) No!! 4>1
- Player 3 change to 1 ⇒ Payoff = (2, 0, 1) No!! 4>1

So (4, 4, 4) is pure NE!!

Answer : The game has two pure Nash Equilibrium

- (I = 1, II = B, III = 1) = (3, 2, 2)
- (I = 2, II = A, III = 2) = (4, 4, 4) #

Question 2 (a)

(I) Describe an incentive compatible mechanism

⇒ The mechanism is the "reverse" Vickrey Auction $\begin{cases} \text{procurement} \\ \text{lowest win} \\ \text{winner pay the second low bid} \end{cases}$

Why use it? Because ~~lo~~ people tell the truth

⇒ means second-price !!

So a standard incentive compatible mechanism for a procurement auction is the Reverse Vickrey Auction:

1. Each seller i submits a bid c_j'.
2. The buyer selects the seller $j = \arg\min_i c_i'$ with the lowest bid
3. The winning seller is paid the second lowest bid, denoted k_j = second-lowest $(c_1', \dots, c_n')$
4. All other sellers receive 0.

The payoff to the winner is: $p_j = k_j - c_j$

In the mechanism, truthful reporting $c_i' = c_i$ is a weakly dominant strategy for all sellers, so the mechanism is incentive compatible. Q.E.D.

(II) Formal proof of incentive compatibility.

Prove that truthful reporting $c_i' = c_i$ is a weakly dominant strategy for every sellers i.

Case 1: Seller i loses when bidding truthfully ⇒ If i bid c_i and doesn't win, then $p_i = 0$.

For any deviation c_i' $\begin{cases} \text{If } c_i' \text{ doesn't become the lowest bid} \to i \text{ still loses} \to \text{payoff} = 0. \\ \text{If } c_i' \text{ become the lowest bid} \to \text{payoff is } p_i = k - c_i \end{cases}$

which may be negative if the second-lowest bid $k < c_i$. Thus deviating can't improve the payoff.

Case 2: Seller i wins when bidding truthfully ⇒ Then payoff is $p_i = k - c_i$ where k is the second-lowest bid and doesn't depend on i's own bid.

(a) Overbidding $c_i' > c_i$

Seller i $\begin{cases} \text{Outcome loses} \to \text{payoff drops from } k - c_i \text{ to } 0 \\ \text{still win} \to \text{payoff unchange} \end{cases}$

(b) Underbidding $c_i' < c_i$

Seller i still wins and receives payment k, payoff $k - c_i$

But underbidding can strictly reduce payoff

Conclusion: Truthful bidding weakly dominates every deviation

Therefore, the Reverse Vickrey Auction is incentive compatible QED.

Question 2 (b)

4 machine 2 of its speed = 1 4 job weight all = 3
2 of its speed = 2

(ii) 1° Consider a symmetric mixed-strategy Nash Equilibrium in which each job independently randomises over the 4 machines.

Let $\begin{cases} p = \text{probability of choosing each "slow" machine } M_1, M_2 \text{ (speed 1)} \\ q = \text{probability of choosing each "fast" machine } M_3, M_4 \text{ (speed 2)} \end{cases}$

$\Rightarrow \quad 2p + 2q = 1 \Rightarrow q = \frac{1}{2} - p$

2° Expected costs ⇒ If a job selects a machine, the other 3 job independently also choose it with probability x giving expected load 9x. (3×3=9)

① If choose slow machine : $E[C_{slow}] = 3 + 9p$ ② Fast machine : $E[C_{fast}] = \frac{3+9q}{2}$

Nash equilibrium condition ⇒ setting them equal ⇒ $3 + 9p = \frac{3+9q}{2} \Rightarrow \begin{cases} p = \frac{1}{18} \\ q = \frac{4}{9} \end{cases}$

Final equilibrium strategy

M_1, M_2 probability $= \frac{1}{18}$ M_3, M_4 probability $= \frac{4}{9}$

3° Formal Proof

① Slow machine : $E[C_{slow}] = 3 + \frac{1}{18} \cdot 9 = 3.5$ ② Fast machine : $E[C_{fast}] = \frac{3 + 9 \cdot \frac{4}{9}}{2} = 3.5$

Since all machine chosen with positive probability give equal expected cost, no job can improve by change. So : <u>This randomised allocation is a Nash equilibrium.</u> ✗ Q.E.D.

(iii) 1° Disprove the statement by providing a counterexample ⇒ Minimum possible makespan

Total work = 4×3 = 12 Total speed = 1+1+2+2 = 6 The theoretical lower bound is $\frac{12}{6} = 2$

But indivisible jobs imply the smallest achievable makespan is 3

2° An optimal allocation

Machine	Jobs	Load	Time
M_1	J_1	3	3
M_2	J_2	3	3
M_3	$J_{3,4}$	3	3
M_4	×	0	0

Makespan = 3

3° Not a Nash equilibrium

. J_1 on M_1 ⇒ Cost = 3

If change to M_4 (speed = 2)

$\frac{3}{2} = 1.5 < 3$

⇒ Want to change

4° Conclusion

The allocation reach minimum makespan but not a Nash Equilibrium.

⇒ <u>The statement is False.</u>

✗ Q.E.D.

I \ II	Costa	Starbucks
Costa	z_{11} (3, 1)	z_{12} (0, 0)
Starbucks	z_{21} (0, 0)	z_{22} (1, 3)

Question 3

(I) $z_{11} = \frac{1}{3}$, $z_{12} = 0$, $z_{21} = 0$, $z_{22} = \frac{2}{3}$

The distribution assigns positive probability only to the two coordination outcomes (Costa, Costa), (Starbucks, Starbucks)

Whenever a player receive a recommendation, they know with certainty what the other player is doing. and following the recommendation always gives them their highest possible payoff in that row or column.

- If Player I is told "Costa" ⇒ they know outcome (Costa, Costa), giving payoff 3 ⇒ no change
- If Player I is told "Starbucks" ⇒ they know outcome (Starbucks, Starbucks), giving payoff 0 ⇒ no change.

⇒ Same as Player II So no player will change to get higher payoff that is correlated equilibrium ※ Q.E.D

(II) Player I constraints If I is recommended
$$\begin{cases} \text{Costa}: 3z_{11} + 0z_{12} \geq 0z_{11} + 1z_{12} \\ \text{Starbucks}: 0z_{21} + 1z_{22} \geq 3z_{21} + 0z_{22} \end{cases} \Rightarrow \begin{cases} 3z_{11} \geq z_{12} \\ z_{22} \geq 3z_{21} \end{cases}$$

Player II constraints If II is recommended
$$\begin{cases} \text{Costa}: 1z_{11} + 0z_{21} \geq 0z_{11} + 3z_{21} \\ \text{Starbucks}: 0z_{12} + 3z_{22} \geq 1z_{12} + 0z_{22} \end{cases} \Rightarrow \begin{cases} z_{11} \geq 3z_{21} \\ 3z_{21} \geq z_{12} \end{cases}$$

Final Constraints ① $3z_{11} \geq z_{12}$ ② $z_{22} \geq 3z_{21}$ ③ $z_{11} \geq 3z_{21}$ ④ $3z_{21} \geq z_{12}$ ⑤ $z_{11} + z_{12} + z_{21} + z_{22} = 1$

⑥ $z_{ij} \geq 0$ for all i, j. ※

(III) Consider the uniform distribution ⇒ $z_{11} = z_{12} = z_{21} = z_{22} = \frac{1}{4} = 0.25$

Check Player I constraint when recommended Starbucks ⇒ $0z_{21} + 1z_{22} \geq 3z_{21} + 0z_{22}$

⇒ $0(0.25) + 1(0.25) \geq 3(0.25) + 0(0.25)$ ⇒ $0.25 \geq 0.75$ No!

This means if Player I is recommended Starbucks, they would prefer to deviate to Costa to get a higher expected payoff. Therefore, it is no correlated equilibrium. ※ Q.E.D.